

Assignment 5

Token Ring Mutual Exclusion Viva Questions

1. What is mutual exclusion?

Mutual exclusion is a technique that allows only one process to enter the critical section at a time.

2. What is a critical section?

A critical section is a part of the program where shared resources are accessed.

3. Why is mutual exclusion needed?

To prevent race conditions and data inconsistency.

4. What is a race condition?

A race condition occurs when multiple processes access shared data simultaneously causing incorrect results.

5. What is Token Ring Algorithm?

Token Ring Algorithm is a distributed mutual exclusion technique where a unique token circulates among processes.

6. What is a token?

A token is a special message that grants permission to enter the critical section.

7. Which process can enter the critical section?

Only the process possessing the token can enter the critical section.

8. What happens after a process completes execution?

It passes the token to the next process.

9. What is the main advantage of token-based algorithms?

They guarantee mutual exclusion using a unique token.

10. What is the role of sequence number?

Sequence numbers distinguish old requests from new requests.

11. What are the requirements of mutual exclusion algorithms?

- No deadlock
 - No starvation
 - Fairness
 - Fault tolerance
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12. What is deadlock?

Deadlock occurs when processes wait forever for resources.

13. What is starvation?

Starvation means a process waits indefinitely to enter the critical section.

14. What is fairness in mutual exclusion?

Every process gets a fair chance to execute the critical section.

15. What is fault tolerance?

The system continues functioning even after failures.

16. Why are shared variables not used in distributed systems?

Because distributed systems do not have shared memory.

17. What is message passing?

Message passing is communication between processes using messages.

18. What are the approaches to distributed mutual exclusion?

1. Token-based approach
 2. Non-token-based approach
 3. Quorum-based approach
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19. Give examples of token-based algorithms.

Suzuki-Kasami Broadcast Algorithm.

20. What is Lamport's Algorithm?

It is a non-token-based mutual exclusion algorithm using timestamps.

21. What is Maekawa's Algorithm?

It is a quorum-based mutual exclusion algorithm.

22. What is the purpose of ServerSocket in Java?

ServerSocket waits for client connection requests.

Example:

```
ServerSocket ss = new ServerSocket(7000);
```

23. What is the purpose of Socket class?

It establishes communication between client and server.

24. Why is multithreading used in the server?

To handle multiple clients simultaneously.

25. Which port number is used in the program?

7000

26. What does Runnable interface do?

It allows a class to execute code in a separate thread.

27. What is the role of BufferedReader?

It reads input data efficiently.

28. What is the role of PrintStream?

It sends output data through sockets.

29. What is the final result of this practical?

Successfully communicated with multiple client machines from the server.

30. What is the outcome of this practical?

A site possessing the unique token is allowed to enter the critical section.
